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A shared-control ultrasound-guided robotic system for liver puncture under respiratory motion

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Ultrasound (US)-guided liver puncture is widely used in diagnosis and minimally invasive intervention. However, the procedure is highly dependent on the surgeon's experience and is substantially affected by respiratory motion (RM), which can reduce imaging stability and puncture accuracy. To address these challenges, this paper presents a shared-control robotic system for US-guided liver puncture under RM. The proposed system integrates a puncture execution mechanism (PEM) that enables coordinated US probe manipulation and needle operation while preserving the surgeon's control over needle orientation and insertion. To improve procedural stability, a dual-stage pose-force assistance strategy is developed. In the first stage, pose-guided assistance is used to position the US probe and obtain a suitable imaging view. In the second stage, force-regulated interaction control is applied to maintain stable probe-skin contact during RM, while the surgeon remains responsible for the puncture action. In addition, a bionic respiratory motion device (BRMD) was developed to simulate thoracoabdominal anatomy and RM using medical-imaging-compatible anthropomorphic phantoms. In phantom experiments, the proposed shared-control system reduced the average operation time, decreased the puncture RMS error, and increased the success rate compared with manual operation. An in vivo study in a Labrador dog was further conducted as a feasibility demonstration.

Keywords Liver puncture, Surgical robot, Shared control, Respiratory motion, Bionic RM device

Liver cancer is one of the most lethal malignancies worldwide, and early diagnosis combined with precise treatment is critical for improving patient survival rates[1]. In clinical practice, percutaneous puncture plays a key role in liver puncture and interventional therapy, typically relying on medical image guidance for accurate targeting and execution[2–5]. Among various imaging modalities, US, owing to its real-time capability, absence of ionizing radiation, and low cost, has become the most commonly used guidance method in liver tumor diagnosis and treatment[6,7]. However, US-guided liver puncture is highly dependent on the surgeon's experience and is particularly is susceptible to RM. RM induces continuous displacement and deformation of the liver and surrounding tissues[8,9], leading to target position variation, degraded imaging stability, and uncertainty in probe–tissue interaction, which in turn reduces puncture accuracy and efficiency and increases the risk of complications such as mispuncture and bleeding.

To mitigate RM-induced disturbances, robotic techniques have been introduced to improve the stability and efficiency of liver puncture, leading to various fully autonomous or strongly model-driven robotic systems[10–12]. These approaches typically rely on explicit models of RM, tissue mechanics, and needle–tissue interaction to predict and plan the puncture process. Although promising results have been reported under controlled experimental conditions, their clinical applicability remains limited due to stringent requirements on system calibration accuracy, imaging stability, and respiratory predictability. Moreover, liver puncture is inherently a tightly coupled perception–decision–execution process, in which surgeons must continuously interpret real-time US images and dynamically adjust their strategies[13]. Consequently, fully autonomous or strongly model-driven robotic robotic solutions still face practical challenges in safety, clinical acceptance, and responsibility attribution under current clinical and technical conditions[14].

From a clinical task perspective, it is therefore necessary to decompose the US-guided liver puncture procedure and to allocate control authority rationally between the robot and the surgeon. Specifically, probe positioning, pose maintenance, and probe–tissue interaction stabilization demand high precision and repeatability and are

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highly sensitive to RM, whereas puncture timing determination, fine trajectory adjustment, and risk avoidance rely heavily on the surgeon's real-time image perception and clinical experience. Under this rationale, shared control has emerged as an important paradigm in medical robotics, whereby the robot assists with precision- and stability-critical subtasks, while the surgeon retains control over perception-driven and critical decision-making operations[15–17].

Some studies have explored this direction. Human-robot collaborative US systems based on teleoperation have demonstrated the feasibility and safety of surgeon-dominant, robot-assisted workflows in puncture scenarios[18]. Semi-automatic puncture systems have also been reported, in which subtasks such as target localization or needle insertion are delegated to robots under strong modeling and registration constraints[19]. In the broader field of interventional robotics, shared-control concepts have been systematically discussed to enable complementary cooperation between robotic systems and clinicians[20]. Nevertheless, systematic investigations that jointly address US probe manipulation stability under RM and its integration with puncture workflows remain limited.

In addition to control strategies, experimental validation under RM presents further challenges. Most existing liver phantoms are static and fail to reproduce respiratory-induced dynamics[21]. Although animal experiments provide higher physiological fidelity, they are associated with high cost, complex ethical approval processes, and limited repeatability[22,23]. Therefore, developing a reproducible, controllable, and physiologically relevant RM experimental platform is essential for systematic evaluation of robot-assisted puncture strategies[24].

Motivated by these considerations, this paper proposes a shared-control US-guided liver puncture robotic system under RM. A PEM is designed to enable coordinated manipulation of the probe and needle within a compact structure, while preserving the surgeon's authority over needle orientation and insertion. A dual-stage pose-force assistance scheme is further developed to establish a favorable imaging configuration through pose guidance and to stabilize the probe-tissue interaction via force regulation, thereby mitigating RM-induced disturbances. In addition, a bionic BRMD is developed for system validation, and comprehensive evaluations including positioning accuracy tests, force-control assessments, phantom experiments, and in vivo Labrador dog experiments are conducted. The specific contributions are summarized as follows:

1) A PEM is designed, which enables coordinated manipulation of the US probe and the puncture needle within a compact structure through an efficient and integrated mechanical design.
2) A dual-stage pose-force collaborative assistance scheme oriented to RM is proposed. By decoupling spatial alignment of the probe from contact stabilization, the method establishes a favorable imaging configuration and regulates the probe-tissue interaction force, thereby effectively enhancing imaging stability and the robustness of shared control under RM.
3) A BRMD is developed, and comprehensive validation is conducted through positioning accuracy tests, force-control evaluations, phantom-based experiments, and in vivo animal experiments on a Labrador dog.

The rest of this paper is organized as follows. The section "System design and mechanical analysis" describes the system structure and mechanism design. The section "Methods" describes the relevant methods. The section "Experimental setup and validation" verifies the developed system and devices. The section "Conclusion and discussion" concludes and discusses this paper.

System design and mechanical analysis

System structure and workflow

As shown in Fig. 1, the proposed US-guided puncture system is designed under a shared-control paradigm and consists of an US imaging device (ACUSON S2000, Siemens Healthineers, Germany), an optical tracking system (OTS; Guangzhou Aimooe Technology Co., Ltd.), a robotic manipulator (UR5e, Universal Robots, Denmark), a PEM, a software control center (SCC), and a lesion target area. The US device provides real-time visualization of internal tissues, which serves as the primary information source for the surgeon during puncture. The SCC integrates CT/US imaging, optical tracking, and robotic state information to support surgeon-in-the-loop

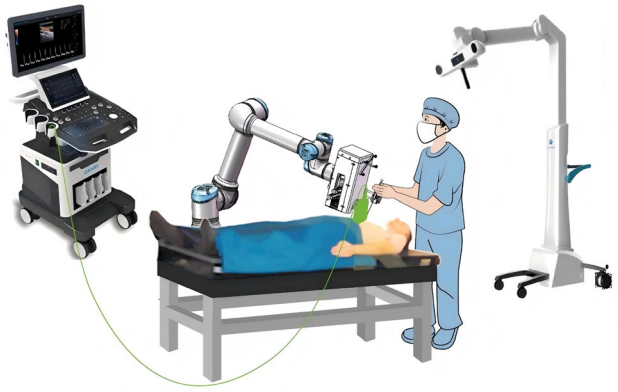


Fig. 1. The shared-control US-guided puncture robotic system structure.

guidance and coordination. The OTS performs spatial tracking and registration among the imaging, robot, and patient coordinate frames, enabling consistent spatial alignment throughout the procedure. The PEM provides robotic assistance for US probe manipulation and image stabilization, while needle insertion remains under direct surgeon control.

The overall workflow is organized into two sequential assistance phases. In the first phase, an OTS-guided pose assistance strategy is employed to guide the robotic manipulator in aligning the US probe with the target region and approaching the planned skin entry site. This phase establishes a stable imaging configuration for the surgeon, rather than executing puncture actions. In the second phase, once probe-skin contact is established, the system transitions to an force-feedback assistance mode to regulate probe-tissue interaction forces and suppress RM-induced disturbances, thereby maintaining imaging stability. Throughout the entire procedure, control authority is explicitly shared between the robotic system and the surgeon: the robot autonomously stabilizes the imaging workspace, while the surgeon performs needle orientation adjustment and insertion based on real-time feedback from both the SCC and US imaging.

Design of the PEM and mechanical motion analysis

The PEM in this study is developed to provide stable support, RM compensation, and force-feedback regulation during the procedure, while leaving the final needle insertion to the surgeon. This design not only meets current clinical requirements for safety and responsibility allocation, but also leverages the surgeon's experience and real-time judgment, balancing the intelligence of robotic assistance with the flexibility of surgeon dominance. Such a strategy lays a practical foundation for the clinical translation of the system. In clinical puncture procedures, it is essential to ensure that the needle can be freely adjusted within 2D plane of the US probe. Therefore, in Fig. 2a, the PEM is equipped with US probe, puncture needle, drive motors, and various motion units. This PEM possesses multi-DOFs, with three of them dedicated to adjusting the spatial pose of the needle within the plane of the US probe. Another DoF is used to adjust the needle so that it is in the plane of the US probe. The final DoF facilitates the guidance and insertion of the needle. In Fig. 2b, along the guide rail of the needle, the needle can move along the X-axis of US probe coordinate system, which is denoted as p_1. When the needle is visualized within the US imaging plane, precise movements of the needle can be achieved through motor control. Specifically, the movement of Motor A realizes vertical motion p_2 of the needle along the Z-axis of the US probe, which is the normal vector of US probe. While the movement of Motor B enables rotation p_3 of the needle around the remote center of motion point within the plane. The coordinated movement of Motor B and Motor C can achieve horizontal motion p_4 of the needle along the Y-axis. Once the spatial position and angle of the needle are adjusted to the optimal pose, the surgeon can perform the puncture along the needle guidance device, represented as p_5.

The motion principle of the PEM is illustrated in Fig. 2c. Let N_a, N_b, and N_c denote the rotation inputs of Motors A, B, and C, respectively, expressed in revolutions. Let R_a, R_b, and R_c denote the three reducers connected to Motors A, B, and C, with corresponding reduction ratios of i_a, i_b, and i_c, respectively. Motor A drives the lead-screw-nut mechanism through reducer R_a. After reduction, one motor revolution produces a screw rotation of $1/i_a$ revolution, and the corresponding linear displacement is determined by the lead-screw pitch p_a. Therefore, the vertical displacement generated by Motor A is $x_a = p_a \cdot N_a/i_a$, which is used to adjust the needle position along the vertical direction within the US imaging plane. Motors B and C drive two rack-and-pinion mechanisms through reducers R_b and R_c, respectively. Thus, the rack displacements driven by Motors B and C can be expressed as $x_b = \pi \cdot d_b \cdot N_b/i_b$ and $x_c = \pi \cdot d_c \cdot N_c/i_c$, where d_b and d_c are the pitch diameters of the corresponding pinion gears. When $x_b = x_c$, the two rack sliders move synchronously, resulting in a horizontal translation of the needle within the US imaging plane. When $x_b \neq x_c$, the differential displacement between the two rack sliders generates an in-plane rotation of the needle around the equivalent RCM. Based on the above transmission relationships, the motor-to-displacement mapping can be written as:

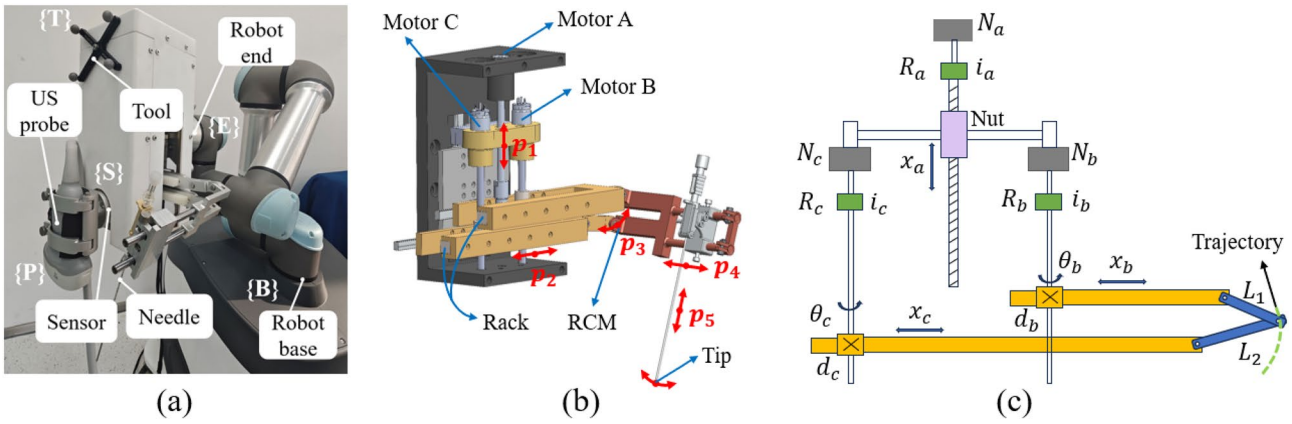


Fig. 2. Description for liver puncture execution mechanism. **a** Components of the mechanism. **b** DoFs of the mechanism. **c** The motion principle of the mechanism.

$$\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} \frac{p_a}{i_a} & 0 & 0 \\ 0 & \frac{\pi d_b}{i_b} & 0 \\ 0 & 0 & \frac{\pi d_c}{i_c} \end{bmatrix} \begin{bmatrix} N_a \\ N_b \\ N_c \end{bmatrix}, \tag{1}$$

where p_a is the lead-screw pitch, d_b and d_c are the pitch diameters of the pinion gears, and i_a, i_b, and i_c are the corresponding reduction ratios. Then, Eq. (1) can be simplified as:

$$\boldsymbol{X} = \boldsymbol{P}\boldsymbol{N}, \tag{2}$$

where $\boldsymbol{P}$ represents the transmission parameter matrix, $\boldsymbol{N}$ represents the motor rotation input vector, and $\boldsymbol{X}$ represents the corresponding linear displacement output vector.

Methods

This study proposes a shared-control US-guided liver puncture paradigm in which control authority is distributed between the robot and the surgeon. The robot focuses on establishing and maintaining a favaroble and satable US imaging under RM, while the surgeon remains in the loop to perform perception-driven decision making and needle insertion based on real-time US feedback. Within this paradigm, a dual-stage control scheme is implemented. In the first stage, pose-guided probe positioning is provided to align the probe with the target region and establish a favorable imaging configuration under RM. In the second stage, force-regulated interaction stabilization is employed to maintain stable probe-skin contact and suppress RM-induced disturbances at the imaging interface. Together, these stages assist the surgeon by enhancing imaging quality and interaction robustness, without altering conventional clinical workflows or compromising the surgeon's leading role during puncture.

Pose-guided US probe positioning under RM

This stage provides pose guidance for US probe positioning under RM and serves as a preparatory step for subsequent shared-control puncture execution. The primary objective is to guide the US probe to a planned body-surface pose that enables stable and clinically usable lesion visualization, while leaving needle orientation and insertion fully under surgeon control.

To ensure geometric consistency and numerical robustness when handling coupled rotation and translation, probe pose estimation and registration are formulated within a unified rigid-body representation. Multiple coordinate frames are introduced to describe the spatial relationships among system components. Specifically, the OTS frame is denoted as O and serves as the global reference; the robot end-effector frame is denoted as E ; the optical tracking tool frame mounted on the PEM is denoted as T; the US probe frame is denoted as P; and the SCC image frame is denoted as I. Through system calibration, a transformation chain from the image space to the OTS frame is established, enabling global localization of the planned imaging target with respect to the physical workspace. As illustrated in Fig. 3a, the pose-guided framework is organized into three nested layers that progressively reduce spatial uncertainty during probe approach: L_1: Skin-mounted markers provide a RM-consistent spatial reference, enabling robust mapping between the planned target region and the OTS frame; L_2 : The lesion target, PEM, and US probe are localized within a unified three-dimensional workspace, ensuring consistent spatial relationships among system components; L_3: The robot end-effector is tracked in real time, and the probe pose is updated online to mitigate RM-induced drift and residual registration mismatch during probe positioning. Together, these layers establish a stable pose reference for subsequent force-regulated imaging stabilization and surgeon-in-the-loop puncture execution. The specific calibration and solution procedure is described as follows.

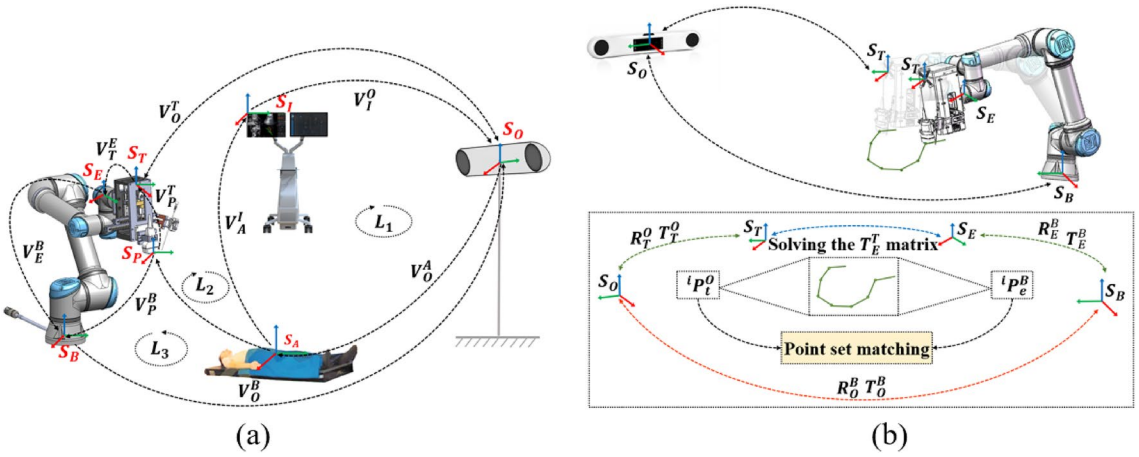


Fig. 3. The pose-guided US probe positioning algorithm schematic diagram. **a** The algorithm framework. **b** Solving transformation relationship $\boldsymbol{R}_O^B$ and $\boldsymbol{T}_O^B$.

Solving the translation matrices $\boldsymbol{T}_E^T$ and $\boldsymbol{T}_E^O$

To establish the spatial relationship between the robot end-effector, the tracking tool, and the OTS frame, a tip-rotation (pivot) calibration procedure commonly adopted in surgical navigation systems is successfully employed[25,26]. In this procedure, the center of the robot end-effector is treated as a fixed pivot point. The robot drives the PEM to perform *n* rotational motions about this pivot while maintaining the end-effector position fixed. Under this geometric constraint, the origin of the tracking tool frame rotates on a concentric spherical surface whose center coincides with the robot end-effector center. During each rotation, the pose information of both the robot end-effector and the tracking tool is recorded. Let the OTS frame be denoted as S_O, the robot end-effector frame as S_E, and the tracking tool frame as S_T. The position of the robot end-effector center expressed in S_T and S_O are denoted as $\boldsymbol{P}_e^T$ and $\boldsymbol{P}_e^O$, respectively, which can be written as:

$$\begin{cases} \boldsymbol{P}_e^T = [{}^1\boldsymbol{P}_e^T, {}^2\boldsymbol{P}_e^T, \ldots, {}^n\boldsymbol{P}_e^T]^T, \\ \boldsymbol{P}_e^O = [{}^1\boldsymbol{P}_e^O, {}^2\boldsymbol{P}_e^O, \ldots, {}^n\boldsymbol{P}_e^O]^T. \end{cases} \tag{3}$$

Then, according to the rigid-body transformation relationship between frames S_O and S_T, the following equations hold:

$$\begin{cases} \boldsymbol{P}_e^T = \boldsymbol{R}_O^T \boldsymbol{P}_e^O + \boldsymbol{T}_O^T, \\ \boldsymbol{P}_t^T = \boldsymbol{R}_O^T \boldsymbol{P}_t^O + \boldsymbol{T}_O^T, \end{cases} \tag{4}$$

By stacking the measurements obtained from *n* different rotational configurations, Eq. (4) can be expanded into a linear system:

$$\begin{pmatrix} {}^1\boldsymbol{R}_T^O & -\boldsymbol{I}_{3\times3} \\ \vdots & \vdots \\ {}^n\boldsymbol{R}_T^O & -\boldsymbol{I}_{3\times3} \end{pmatrix} \begin{pmatrix} \boldsymbol{P}_e^T \\ \boldsymbol{P}_e^O \end{pmatrix} = \begin{pmatrix} -{}^1\boldsymbol{P}_t^O \\ \vdots \\ -{}^n\boldsymbol{P}_t^O \end{pmatrix}, \tag{5}$$

where ${}^i\boldsymbol{P}_t^O$ and ${}^i\boldsymbol{R}_T^O$ are directly obtained from the OTS. The unknown vectors $\boldsymbol{P}_e^T$ and $\boldsymbol{P}_e^O$ are solved using a least-squares approach. Based on the resulting geometric relationships, the translation matrices $\boldsymbol{T}_E^T$ and $\boldsymbol{T}_E^O$ can then be determined.

Solving the transformation $\boldsymbol{R}_O^B$ and $\boldsymbol{T}_O^B$

To establish a consistent spatial relationship between the OTS and the robotic arm base, a point-set registration procedure is performed. During robot motion, the tracking tool mounted on the PEM is continuously observed by the OTS, while the position of the robot end-effector center is simultaneously obtained from the robot. This process yields two corresponding point sets expressed in the OTS frame S_O and the robot base frame S_B, respectively, as illustrated in Fig. 3b. The center point of the robot end-effector is selected as an intermediary reference between the two coordinate frames. By driving the robot to *n* distinct poses and recording the end-effector center position at each pose, the corresponding point sets are given by $\{{}^1\boldsymbol{P}_e^O, {}^2\boldsymbol{P}_e^O, \ldots, {}^n\boldsymbol{P}_e^O\}^T$ and $\{{}^1\boldsymbol{P}_e^B, {}^2\boldsymbol{P}_e^B, \ldots, {}^n\boldsymbol{P}_e^B\}^T$, where the relationship between the two frames satisfies ${}^i\boldsymbol{P}_e^B = \boldsymbol{R}_O^B\, {}^i\boldsymbol{P}_e^O + \boldsymbol{T}_O^B$. Based on these point correspondences, the covariance matrix $\boldsymbol{H}_O^B$ is constructed as:

$$\boldsymbol{H}_O^B = \sum_{i=1}^{n} \left({}^i\boldsymbol{P}_e^B - \frac{1}{n}\sum_{j=1}^{n} {}^j\boldsymbol{P}_e^B \right) \left({}^i\boldsymbol{P}_e^O - \frac{1}{n}\sum_{j=1}^{n} {}^j\boldsymbol{P}_e^O \right)^T, \tag{6}$$

where, the rotation matrix $\boldsymbol{R}_O^B$ is obtained via singular value decomposition (SVD) of $\boldsymbol{H}_O^B$[27]. Subsequently, the translation vector $\boldsymbol{T}_O^B$ is computed as:

$$\boldsymbol{T}_O^B = \frac{1}{n}\sum_{i=1}^{n} {}^i\boldsymbol{P}_e^O - \boldsymbol{R}_O^B \left(\frac{1}{n}\sum_{i=1}^{n} {}^i\boldsymbol{P}_e^B \right). \tag{7}$$

To improve robustness against measurement noise and pose-dependent fluctuations, the initially estimated transformation is further refined. Due to tracking noise and robot positioning uncertainty, the rotation matrices $\boldsymbol{R}_O^B$ obtained from different pose configurations may exhibit small variations. To mitigate this effect, each rotation matrix is first converted into Euler angles $(\alpha_i, \beta_i, \gamma_i)$, and the final optimized rotation is reconstructed using their averaged values:

$$\boldsymbol{R}_{final} = R\left(\frac{1}{n}\sum_{i=1}^{n}\alpha_i,\ \frac{1}{n}\sum_{i=1}^{n}\beta_i,\ \frac{1}{n}\sum_{i=1}^{n}\gamma_i \right), \tag{8}$$

where $R(\cdot)$ denotes the rotation reconstruction function. Similarly, the translation vectors $\boldsymbol{T}_O^B$ estimated at different poses are averaged to obtain the final translation:

$$\boldsymbol{T}_{final} = \frac{1}{n}\sum_{i=1}^{n} {}^{i}\boldsymbol{T}_{O}^{B}, \tag{9}$$

where, the refined transformation $\{\boldsymbol{R}_{final}, \boldsymbol{T}_{final}\}$ provides a more consistent and reliable spatial mapping between the OTS and robot base frames, which is essential for stable probe positioning under respiratory motion.

Solving $\boldsymbol{R}_{E}^{T}$ and target pose localization

As shown in Fig. 3, the rotation relationship between the robot end-effector and the tracking tool can be solved in a closed-form manner[28]. Ideally, the rotation matrix $\boldsymbol{R}_{T}^{E}$ remains constant across different robot poses. However, due to measurement noise from the OTS and robot encoders, slight variations may occur in practice. To suppress random errors, the final rotation matrix is obtained by averaging over multiple measurements:

$$\boldsymbol{R}_{T}^{E} = \frac{1}{m}\sum_{i=1}^{m} {}^{i}\boldsymbol{R}_{B}^{E} \cdot \boldsymbol{R}_{O}^{B} \cdot {}^{i}\boldsymbol{R}_{T}^{O}. \tag{10}$$

Based on the closed-loop kinematic formulation[28] and the target pose-localization strategy described in[10], the desired robot end-effector pose corresponding to the planned target location can be computed as:

$$\begin{cases} \boldsymbol{R}_{E}^{B} = \boldsymbol{R}_{O}^{B}\boldsymbol{R}_{A}^{O}\boldsymbol{R}_{T}^{P}\boldsymbol{R}_{E}^{T}, \\ \boldsymbol{T}_{E}^{B} = \boldsymbol{R}_{O}^{B}\boldsymbol{R}_{A}^{O}\boldsymbol{R}_{T}^{P}\boldsymbol{T}_{E}^{T} + \boldsymbol{R}_{O}^{B}\boldsymbol{R}_{A}^{O}\boldsymbol{T}_{T}^{P} + \boldsymbol{R}_{O}^{B}\boldsymbol{T}_{A}^{O} + \boldsymbol{T}_{O}^{B}, \end{cases} \tag{11}$$

where, the resulting transformation $\{\boldsymbol{R}_{E}^{B}, \boldsymbol{T}_{E}^{B}\}$ is converted into a six-dimensional pose representation using the Rodrigues formula[29], which can be directly interpreted by the robotic controller to guide probe positioning.

Force-regulated probe-skin interaction stabilization under RM

This stage aims to stabilize the US imaging condition under RM-induced disturbances by regulating probe–skin interaction. We formulate the problem as safety-constrained normal-force regulation with tangential compliance, where the robot autonomously maintains the contact force within a prescribed band to suppress motion-induced image jitter while avoiding excessive compression and shear.

Force baseline compensation and gravity correction

To ensure accurate probe–tissue interaction regulation, the raw force measurements must be compensated for gravitational effects and sensor bias. Let S_w, S_b, S_e, and S_s denote the world, robot base, end-effector, and force sensor coordinate frames, respectively. The force measured by the sensor can be decomposed as:

$$\boldsymbol{F}_{s} = \boldsymbol{F}_{ex} + \boldsymbol{G}_{s} + \boldsymbol{F}_{o}, \tag{12}$$

where $\boldsymbol{F}_{ex}$ is the external contact force acting on the US probe, $\boldsymbol{G}_{s}$ represents the equivalent gravitational force of the probe and its clamping components expressed in S_s, and $\boldsymbol{F}_{o}$ denotes the constant sensor bias. The gravitational component $\boldsymbol{G}_{s}$ depends on the robot configuration and can be expressed as:

$$\boldsymbol{G}_{s} = -\boldsymbol{R}_{b}^{s}\boldsymbol{R}_{w}^{b}\hat{\boldsymbol{G}}_{w}, \tag{13}$$

where $\hat{\boldsymbol{G}}_{w}$ is the gravity vector of the attached load expressed in the world frame, $\boldsymbol{R}_{w}^{b}$ is the rotation from S_w to S_b, and $\boldsymbol{R}_{b}^{s}$ denotes the rotation from the robot base frame to the sensor frame. By collecting force measurements at n distinct robot poses without external contact, the gravity vector and sensor bias can be jointly identified. Stacking the measurements yields the linear system, i.e.,

$$\boldsymbol{F}_{s}^{(n)} = \begin{bmatrix} \boldsymbol{R}_{b}^{s}\boldsymbol{R}_{w}^{b} & \boldsymbol{I} \\ \vdots & \vdots \\ \boldsymbol{R}_{b}^{s}\boldsymbol{R}_{w}^{b} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{G}}_{w} \\ \boldsymbol{F}_{o} \end{bmatrix}, \tag{14}$$

which can be solved using a least-squares approach[30]. Once $\hat{\boldsymbol{G}}_{w}$ and $\boldsymbol{F}_{o}$ are identified, the external contact force $\boldsymbol{F}_{ex}$ can be computed in real time according to Eq. (12), providing a reliable force baseline for subsequent interaction regulation.

Force-regulated interaction stabilization under RM

To suppress RM-induced disturbances during US-guided puncture, a force-regulated interaction stabilization strategy is employed, as illustrated in Fig. 4. The objective of this strategy is to maintain stable probe-skin contact and preserve a favorable US imaging window during RM. After force baseline compensation and gravity correction, the external contact force acting on the US probe can be computed from Eq. (12), which is then transformed into the probe coordinate frame and expressed as:

$${}^{p}\boldsymbol{F}_{ex} = \begin{bmatrix} {}^{p}f_{x}^{ex} & {}^{p}f_{y}^{ex} & {}^{p}f_{z}^{ex} \end{bmatrix}^{T}, \tag{15}$$

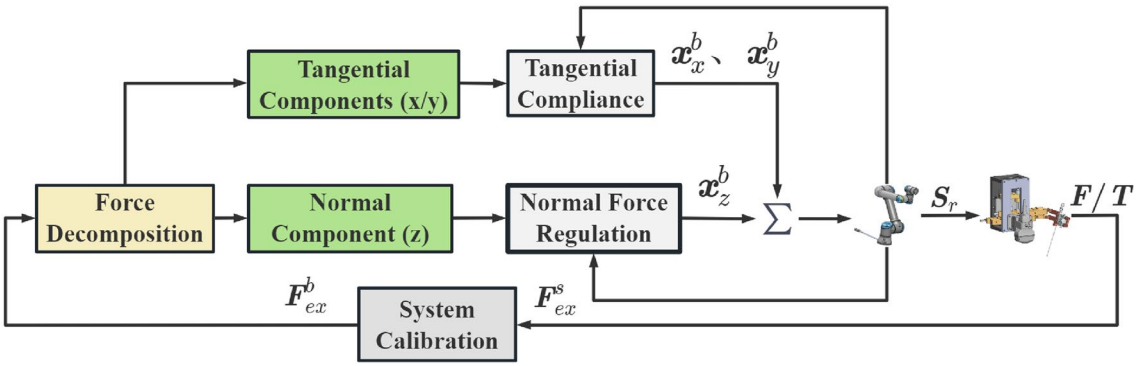


Fig. 4. Overview of the force-regulated interaction stabilization framework under RM.

where ${}^p f_z^{ex}$ denotes the contact force along the probe normal direction, and ${}^p f_x^{ex}$ and ${}^p f_y^{ex}$ denote the tangential interaction forces.

In this study, a direction-dependent force regulation scheme is adopted. Along the probe normal direction, the robot actively regulates the contact force to track a prescribed value f_d, thereby maintaining stable coupling between the probe and the skin. Along the tangential directions, the controller minimizes the contact force as much as possible. Therefore, the desired interaction force is defined as ${}^p\boldsymbol{F}_d = [0, \quad 0, \quad f_d]^T$, and the corresponding force error is given by ${}^p\boldsymbol{e}_f = {}^p\boldsymbol{F}_d - {}^p\boldsymbol{F}_{ex}$. Here, the zero tangential force reference indicates that no active tangential contact force is intentionally applied. It does not imply that the tangential forces are strictly maintained at zero, since residual tangential forces may still exist due to RM-induced deformation, surface friction, and coordinate coupling. Then, the force error is mapped to a smooth Cartesian motion compensation of the probe through an admittance interaction control model:

$$\boldsymbol{M}_d\Delta\ddot{\boldsymbol{x}} + \boldsymbol{B}_d\Delta\dot{\boldsymbol{x}} + \boldsymbol{K}_d\Delta\boldsymbol{x} = {}^b\boldsymbol{e}_f, \tag{16}$$

where $\Delta\boldsymbol{x} = [\Delta x, \Delta y, \Delta z]^T$ denotes the probe motion compensation expressed in the robot base frame S_B. $\boldsymbol{M}_d$, $\boldsymbol{B}_d$, and $\boldsymbol{K}_d$ are the virtual inertia, damping, and stiffness matrices, respectively. For each direction $n \in \{x, y, z\}$, Eq. (16) can be expressed as:

$$\boldsymbol{m}_n^d\Delta\ddot{\boldsymbol{x}}_n + \boldsymbol{b}_n^d\Delta\dot{\boldsymbol{x}}_n + \boldsymbol{k}_n^d\Delta\boldsymbol{x}_n = \boldsymbol{e}_{f,n}^b. \tag{17}$$

This formulation allows the normal contact force to be regulated around the prescribed value f_d, while the tangential directions are treated as compliant directions to reduce excessive shear interaction.

In addition, to discuss the interaction stability with tissues of different stiffness, the local probe-tissue contact can be approximated as a passive linear viscoelastic model:

$${}^b\boldsymbol{F}_{ex} = \boldsymbol{F}_0 + \boldsymbol{K}_e\Delta\boldsymbol{x} + \boldsymbol{B}_e\Delta\dot{\boldsymbol{x}}, \tag{18}$$

where $\boldsymbol{K}_e$ and $\boldsymbol{B}_e$ denote the equivalent tissue stiffness and damping, respectively. A smaller $\boldsymbol{K}_e$ corresponds to a softer contact condition, whereas a larger $\boldsymbol{K}_e$ corresponds to a stiffer contact condition. Substituting this contact model into Eq. (16) yields the local closed-loop interaction dynamics:

$$\boldsymbol{M}_d\Delta\ddot{\boldsymbol{x}} + (\boldsymbol{B}_d + \boldsymbol{B}_e)\Delta\dot{\boldsymbol{x}} + (\boldsymbol{K}_d + \boldsymbol{K}_e)\Delta\boldsymbol{x} = \boldsymbol{F}_d - \boldsymbol{F}_0. \tag{19}$$

Given the positive diagonal matrices $\boldsymbol{M}_d$, $\boldsymbol{B}_d$, and $\boldsymbol{K}_d$, together with a passive tissue environment, the closed-loop interaction remains locally stable under the considered contact conditions. In practice, bounds on velocity, displacement, and contact force are further imposed to avoid excessive compression or interaction oscillation in relatively stiff regions.

For numerical implementation, the second-order interaction model is rewritten in a first-order state-space form:

$$\begin{cases} \Delta\dot{\boldsymbol{x}}_n = \boldsymbol{v}_n, \\ \dot{\boldsymbol{v}}_n = (\boldsymbol{m}_n^d)^{-1}\left(\boldsymbol{e}_{f,n}^b - \boldsymbol{b}_n^d\boldsymbol{v}_n - \boldsymbol{k}_n^d\Delta\boldsymbol{x}_n\right), \end{cases} \tag{20}$$

then, the state equations are computed using a fourth-order Runge-Kutta (RK4) scheme[31] to generate smooth motion compensation. The compensated probe reference is then given by $\boldsymbol{x}_{ref} = \boldsymbol{x}_0 + \Delta\boldsymbol{x}$. The reference pose $\boldsymbol{x}_{ref}$ is tracked by a PID-based robotic servo controller[10], which can drive the robot to the target position. It should be noted that the direction-dependent regulation in this study refers to different force-control objectives in the normal and tangential directions, rather than anisotropic values of the virtual inertia, damping, and stiffness matrices. Specifically, the normal direction is actively regulated to track the prescribed contact force, whereas the tangential directions are treated with compliant shear reduction. Therefore, isotropic diagonal

matrices $\boldsymbol{M}_d$, $\boldsymbol{B}_d$, and $\boldsymbol{K}_d$ can be used to provide uniform dynamic smoothing and simplify parameter tuning in the current implementation.

Shared-control puncture strategy and puncture execution logic

A shared-control puncture strategy is proposed, in which the robot and the surgeon collaboratively perform complementary subtasks during US-guided liver puncture under RM. Rather than pursuing fully autonomous puncture execution, the proposed strategy aims to establish and maintain a stable puncture environment through robotic assistance, while allowing the surgeon to retain full authority over needle orientation adjustment and insertion timing. Meanwhile, the robotic arm supporting the US probe significantly reduces the burden on surgeons during prolonged probe-holding operations. The overall puncture execution workflow consists of the following three stages.

Preoperative planning and respiratory-gated probe positioning

In the preoperative stage, CT scans of the lesion region are acquired and imported into the SCC for 3D reconstruction, after which the surgeon performs path planning based on clinical experience. During intraoperative execution, the robotic manipulator drives the PEM toward the planned body-surface location using pose-guided control under RM. To mitigate the mismatch between the static CT reference and continuously varying respiratory states, a respiratory gating strategy is employed[10]. Specifically, probe positioning motions are permitted only when the registration error between the image space and the physical space falls below a predefined threshold; when the error exceeds the threshold, probe motion is temporarily suspended until the gating condition is satisfied. Through this mechanism, the robot assists in guiding the US probe to the vicinity of the planned probe pose, thereby establishing an initial imaging configuration for the surgeon. At this stage, the robot does not execute any needle insertion or needle orientation control.

Probe-skin interaction stabilization under RM

Once the US probe reaches the planned body-surface region and establishes contact with the skin, the system transitions to a force-regulated assistance mode. In this mode, the robot stabilizes the probe-skin interaction by regulating the 3D contact force between the probe and the skin. This force regulation strategy does not aim to compensate for internal organ motion or predict respiratory patterns; instead, it focuses on ensuring stable probe contact conditions to maintain adequate US imaging quality. As a result, despite respiration-induced surface motion, the US imaging can remain relatively stable.

Surgeon-in-the-loop puncture execution

Under the stabilized imaging condition, the surgeon performs needle orientation adjustment and insertion using the needle manipulation module integrated into the PEM. The surgeon relies on real-time US feedback to confirm lesion visibility and flexibly determine the needle insertion direction and timing. During this stage, a human-robot shared-control paradigm is explicitly adopted: the robot continuously maintains a stable imaging workspace through force regulation, while the surgeon is responsible for perception, decision making, and puncture execution. During puncture process, residual relative sliding and cyclic image drift may occur and may cause deviation of the actual puncture entry point from the planned location. However, the surgeon can continuously evaluate target visibility, needle-path visibility, and image stability in real time, and perform puncture during a favorable respiratory phase when the target is clearly visualized and the relative motion is relatively small. This collaborative control strategy preserves the surgeon's clinical authority while reducing the influence of RM-induced imaging fluctuation and residual sliding on puncture execution.

Experimental setup and validation

To comprehensively evaluate the proposed shared-control US-guided liver puncture system, a series of experiments are conducted to verify both individual functional modules and overall system performance under RM. Specifically, the experimental study consists of five parts. Firstly, the design and motion fidelity of the BRMD are validated to ensure repeatable and physiologically relevant RM patterns. Secondly, the positioning accuracy of the pose-guided probe positioning stage is quantitatively evaluated. Thirdly, the performance of the force-regulated interaction stabilization strategy is assessed under respiratory disturbances. Fourthly, phantom-based puncture experiments are conducted on the BRMD platform to verify the effectiveness of the proposed shared-control framework under controlled dynamic conditions. Finally, a labrador dog in vivo experiment is performed to evaluate system robustness and procedural safety under physiological RM, supporting clinical relevance.

Design and motion verification of the BRMD

Currently, most liver phantoms are static models that cannot simulate realistic physiological dynamics[21], which presents significant limitations when evaluating the robustness of US-guided puncture assistance under RM. In particular, RM introduces continuous disturbances at the probe–tissue interface, making it necessary to validate whether a shared-control robotic system can maintain stable imaging conditions during dynamic operation. To support such evaluation, we previously developed a simplified RM simulation model[10], in which RM was generated using a mathematical formulation and a volunteer's chest RM data.

Building upon this foundation, a BRMD compatible with US and CT imaging is developed, as shown in Fig. 5. The device is designed as an experimental testbed to reproduce representative thoracoabdominal anatomy and respiratory-induced motion in a controlled and repeatable manner. For organ simulation, different materials are used to represent the lungs, heart, and liver. As illustrated in Fig. 5a, the thoracic structure includes a resin-based sternum, anatomically shaped lungs, a heart cast, and a liver molded from gel material with embedded tumor

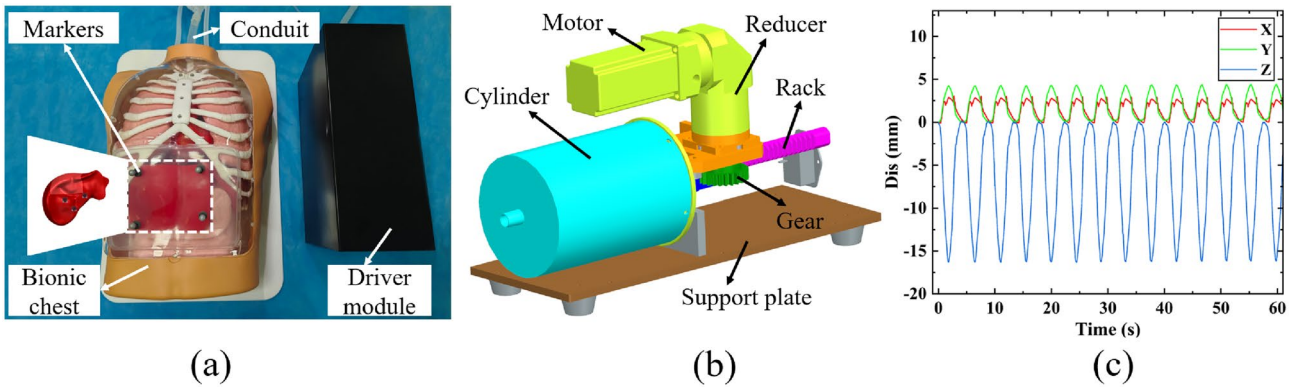


Fig. 5. The liver bionic RM device. **a** Device details with anatomical structure. **b** Schematic diagram of the device motion. **c** Validation of the BRMD motion performance.

targets. The liver surface is covered with silicone-based muscle tissue and an outer transparent skin layer. Four marker spheres are affixed to the skin surface for motion tracking during experiments. All simulated tissues are compatible with medical imaging modalities. In Fig. 5b, the RM drive module, uses motors, gear reducers, and pneumatic transmission to generate lung inflation and deflation according to predefined mathematical models[32], thereby producing repeatable respiratory motion profiles. Specifically, the ventilation volume is modeled as:

$$v(t) = k \cdot i \cdot r \cdot \left[\sum_{n=1}^{N} a_n \cos\left(n\omega t + \varphi_n\right) + b_n \right] \tag{21}$$

where, $v(t)$ denotes the ventilation volume; n is the higher-order frequency factor, representing the number of oscillations or the degree of subdivision; a_n and φ_n are the amplitude and phase of the cosine function, which may vary depending on the respiratory pattern; r is the reduction ratio of the gearbox; i is the ratio of the gear to the rack; k is the mapping coefficient between the rack stroke and the gas volume; t is the time; ω is the angular frequency defined as $\omega = 2\pi f$, where f is the fundamental RM frequency; and b_n represents the initial position. The respiratory pattern can be adapted to various respiratory dynamics by combining cosine functions of different orders.

The respiratory pattern can be adapted to different motion characteristics by combining cosine functions of multiple orders, allowing flexible generation of representative RM profiles for experimental evaluation. The motion performance of the BRMD is then validated. The initial lung gas volume within the thoracic cavity is defined as V_0, which, corresponds to a reference configuration used for system initialization. Subsequently, according to[33], the RM frequency is set to 14 cycles min^{-1}, with a motion amplitude of 10–20 mm per cycle. After parameter initialization, the driving unit is activated to generate respiratory motion of the thoracic phantom. During operation, the spatial positions of surface markers are continuously recorded using the OTS, as shown in Fig. 5c. In this figure, the measured displacements along the X-, Y-, and Z-axes are approximately 3 mm, 5 mm, and 16 mm, respectively, resulting in an overall motion amplitude of about 17 mm with a period of 4.3 s. These results are consistent with reported ranges of human RM and confirm that the BRMD can provide a stable and repeatable dynamic environment for evaluating the robustness of the proposed shared-control puncture assistance system under RM.

Probe positioning accuracy evaluation

To validate the localization accuracy of the proposed pose controller, an experiment is designed based on the closed-loop multi-loop algorithm[28], which does not require any external devices. Instead, it utilizes the closed-loop circuit principle to directly compute the accuracy of the algorithm. Specifically as follows:

$$E_{\text{dis}} = \| \boldsymbol{R}_B^O \cdot (\boldsymbol{R}_E^B \cdot (\boldsymbol{R}_T^E \cdot \boldsymbol{P}_T + \boldsymbol{T}_T^E) + \boldsymbol{T}_E^B) + \boldsymbol{T}_B^O - \boldsymbol{P}_O \| \tag{22}$$

where, $\boldsymbol{P}_T$ and $\boldsymbol{P}_O$ are the corresponding point in the S_O and S_T coordinate systems, respectively. $\boldsymbol{R}_\alpha^\beta$ and $\boldsymbol{T}_\alpha^\beta$ represent the pose relationship between α and β, respectively. α and β are specifically represented by the different coordinate system symbols mentioned above. Specially, the robotic arm is controlled to move the PEM in fixed steps of 6.0 cm per step, completing a total of 27 steps in a cubic pattern, as shown in Fig. 6a. In this figure, the red coordinate system represents the robotic arm's base coordinate system, while the blue coordinate system represents the OTS coordinate system. After each step, the positioning error was recorded, with the maximum error measured at 0.32 mm, the minimum error at 0.15 mm, and an RMS error of 0.21 mm, as shown in Fig. 6b. Compared to existing methods, the proposed approach fully leverages the closure and decoupling properties of the calibration process, achieving lower computational complexity while maintaining high accuracy and robustness. Specifically, the translation pose estimation requires only 12 steps, and the rotation pose estimation requires only 6 steps, significantly reducing the calibration time to less than 1 minutes.

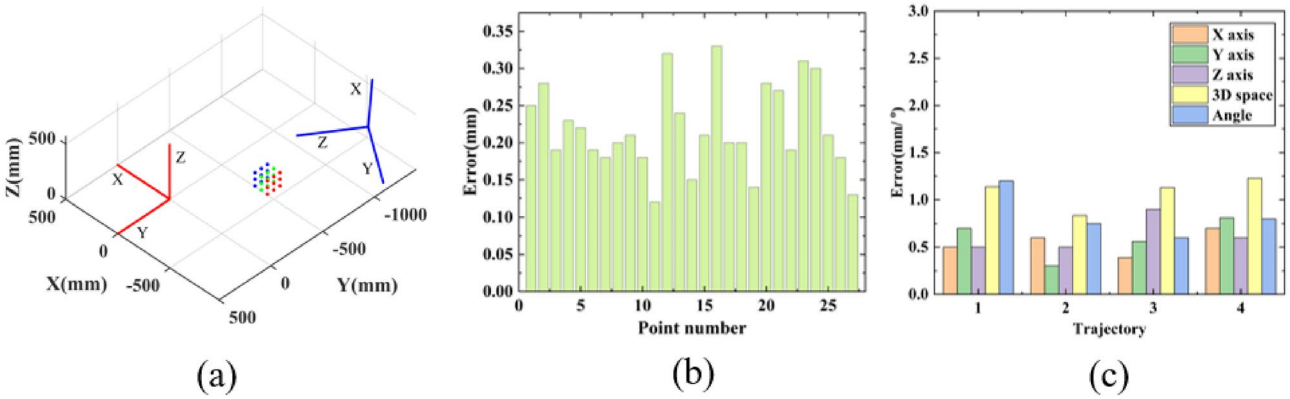


Fig. 6. Calibration experiment and localization validation of the pose controller. **a** Calibration trajectory of the robotic arm in the OTS coordinate system. **b** Statistical analysis of calibration Errors. **c** Error analysis of the localization.

This optimization greatly minimizes data acquisition requirements and computational burden while effectively preventing error accumulation. Experimental results demonstrate that the proposed method outperforms conventional approaches in terms of computational efficiency, calibration accuracy, and system robustness, offering a more efficient, stable, and practical solution for robotic surgical navigation and precise localization.

Subsequently, under the guidance of the OTS, the robotic arm is controlled using the pose controller to drive the PEM to accurately reach the target region, allowing the US probe to contact the skin surface with a specified orientation. To evaluate the accuracy, four different predefined target poses and trajectories are executed. For each one, the robotic system actuated the US probe three times, and the final positioning errors are recorded after each attempt. As shown in Fig. 6c, the horizontal axis represents the index of different target trajectories, while the vertical axis shows the corresponding positioning errors. These include translational errors along the X-, Y-, Z- directions, the overall spatial error computed as the Euclidean distance from the target point, and the angular error representing the deviation in orientation. The units are millimeters (mm) for position errors and degrees ($^\circ$) for angular errors. After quantitative analysis, we can get that the system can achieve an RMS position error of 0.35 mm and an angular error of 0.30°, demonstrating the high precision and repeatability of the proposed pose control strategy.

Force-regulated interaction stabilization verification

Force baseline calibration and gravity compensation verification

Three different robotic motion scenarios are designed to filter gravity bias, including single-axis motion and coupled two-axis motion[34]. A step size of 0.02 rad is set, with a motion range of π. The robot subsequently performs movements in each of these scenarios, recording the data for every spatial point. According to Eq. 14, the target matrix $\boldsymbol{F}_o$ can be solved as $[0.10\,\mathrm{N}, 2.10\,\mathrm{N}, 4.82\,\mathrm{N}]^T$, and the effective equivalent mass can be calculated as $\hat{m} = 0.26\,\mathrm{kg}$. To further evaluate the accuracy of the force bias compensation method, two sets of force data are recorded during the robotic motion. The first set comprises the actual force measurements obtained from the force sensor, serving as the experimental data. The second set represents the theoretical predictions, calculated using the equivalent gravitational force combined with the real-time pose of the robotic arm via Eq. 12. As illustrated in Fig. 7a, the solid lines denote the measured values, while the dashed lines indicate the predicted values. The red, green, and blue curves correspond to the X, Y, and Z force components, respectively. The degree of overlap between the two datasets reflects the effectiveness of the compensation; closer alignment indicates better performance. The average deviation between the measured and predicted forces are approximately 0.12 N, 0.19 N, and 0.11 N, confirming the validity of the compensation method.

Performance evaluation of force-regulated interaction stabilization

In clinical practice, the contact force between the US probe and the human body should be maintained within the range of 2–15 N[35]. Here, a constant force of 6 N along the normal vector of the US probe was first applied, pressing it against the surface of a chest-abdominal cavity model. Subsequently, RM is activated, and the actual forces acting on the US probe during this process are measured, as shown in Fig. 7b. The forces along the X-, Y-, and Z-axes of the US probe's coordinate system exhibited certain rules. The Z-axis component showed the largest amplitude variation, fluctuating between 0 N and 12 N, while the X- and Y-axis components respectively fluctuated within the range of 0-5 N and 0-3 N. Based on the motion amplitude of the target region during the respiratory simulation and its corresponding force effect on the US probe, appropriate impedance characteristic coefficients are selected. After experimental testing and adjustment, the determined values for $\boldsymbol{M}$, $\boldsymbol{B}$, and $\boldsymbol{K}$ are as follows:

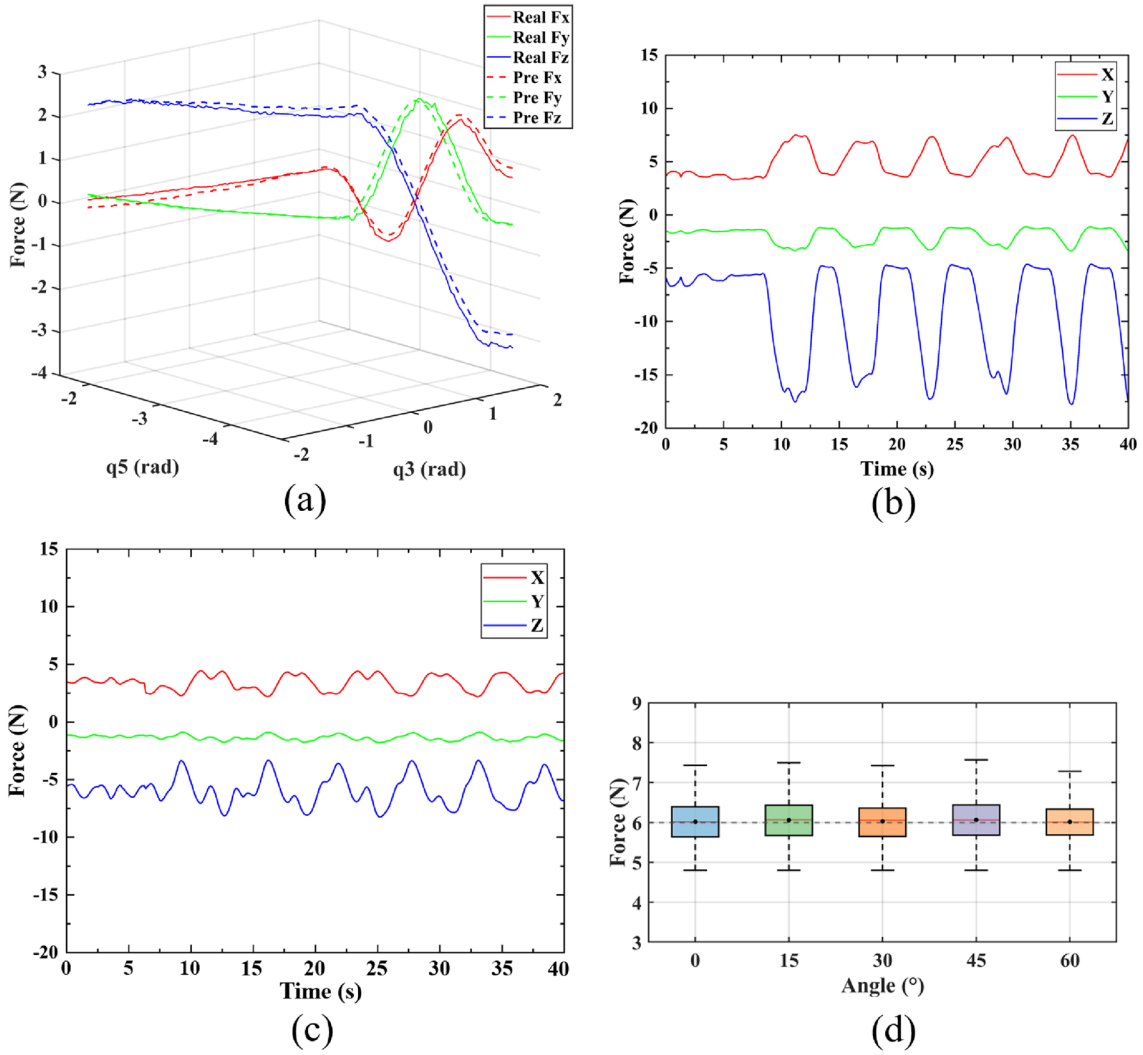


Fig. 7. Verification for compensation control algorithm. **a** Variation curves of measured and predicted values. **b** Force conditions without compensation control algorithm. **c** Force conditions with compensation control algorithm. **d** Force values at different angles.

$$\begin{aligned} \boldsymbol{M} &= \mathrm{diag}[0.5, 0.5, 0.5] \\ \boldsymbol{B} &= \mathrm{diag}[20, 20, 20] \\ \boldsymbol{K} &= \mathrm{diag}[10, 10, 10] \end{aligned} \tag{23}$$

Under the same conditions, the robot is controlled to drive the PEM to compensate for the RM. The actual forces curve acting on the US probe are shown in Fig. 7c. During the initial phase, the system quickly reaches a stable state ensuring that the forces along each axis of the US coordinate system fluctuated within a small range. Where, the Z-axis component oscillated around 6 N, with a fluctuation range of 5-7 N, while the X- and Y-axis components respectively exhibited regular oscillations within 0-3 N. Then, define the angle θ_Z as the absolute angle between the US probe's normal vector and the Z-axis of the robotic arm's base coordinate system. Measurements similar to those in Fig. 7c are performed at different angles, recording the actual forces along the US probe's normal vector during the RM. Based on the definition of θ_Z as the absolute angle between the normal vector of the US probe and the Z-axis of the robotic arm's base coordinate system, additional compensation experiments are conducted under different θ_Z conditions. The actual contact forces along the probe's normal direction during RM qre recorded, and the results are shown in Fig. 7d. The boxplots demonstrate that, regardless of the value of θ_Z, i. e., the contact force consistently remained stable around the prescribed 6 N with only minor fluctuations. In addition, the central horizontal line inside each box represents the median of the measured forces, while the black dot denotes the mean value at each angle. Consistent with the results in Fig. 7c, these findings further verify that the proposed adaptive force-feedback compensation control algorithm can maintain stable and effective performance under varying angular conditions, highlighting its robustness and general applicability.

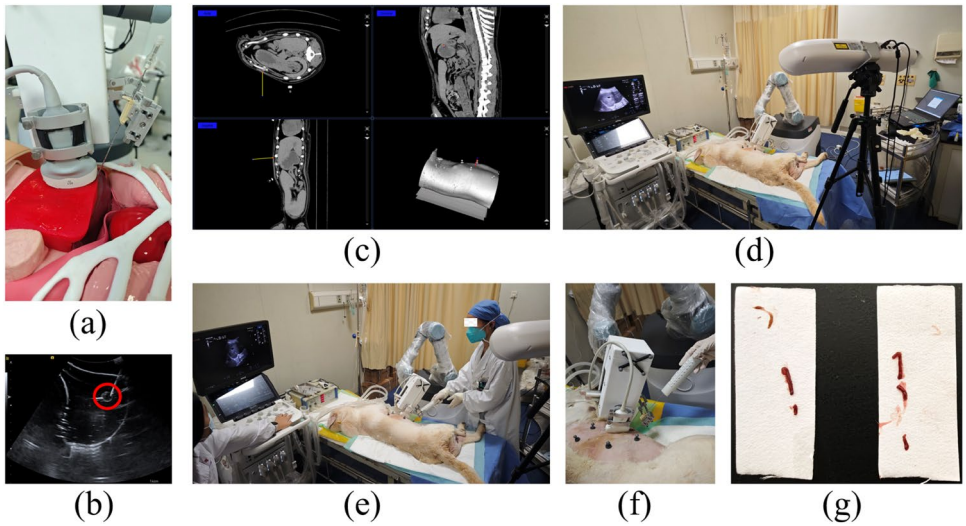


Fig. 8. BRMD and animal experiment verification. (**a**, **b**) Puncture scene and result based on BRMD. (**c**–**g**) Labrador animal experiment scene, where (**c**) 3D Reconstruction and Path Planning, (**d**) Position-guided US probe positioning, (**e**) The surgeon performs puncture, (**f**) The final puncture result scene, and (**g**) Removed liver disease tissue.

BRMD puncture experimental validation

This section presents a dynamic experiment based on the BRMD, aiming to evaluate the preliminary performance of the proposed robotic system in terms of puncture accuracy, success rate, and procedural efficiency under RM disturbances. Firstly, under the control of the hybrid force–pose controller, the robotic arm drives the PEM to the target region on the body surface, allowing the US probe to make stable contact with the skin. A contact force of approximately 6 N is maintained in the normal direction. During this process, the surgeon continuously adjusts the needle orientation and performs the puncture operation based on real-time US imaging. The puncture result visualized in the US image is presented in Fig. 8a, and the corresponding real scene is shown in Fig. 8b. The surgeon then manually inserts the needle along the trajectory under real-time US image guidance. To assess the performance of the system, three different target regions are selected. For each target, the surgeon performs three punctures manually and three with robotic assistance. In this limited phantom comparison, the robot-assisted method showed improved performance across the evaluated metrics. Given the limited sample size, the following statistical analysis was treated as exploratory rather than confirmatory. Operation time and RMS targeting error were calculated only from successful punctures, whereas the success rate was calculated from all puncture attempts. Specifically, for successful punctures, the mean procedure time decreased from 2.98 ± 0.57 min to 2.03 ± 0.28 min, and the mean RMS targeting error decreased from 2.73 ± 0.77 mm to 1.31 ± 0.27 mm. Exploratory Welch's t-tests indicated differences in procedure time ($p = 0.039$) and RMS targeting error ($p = 0.031$). The success rate increased from 44.4% (4/9) in manual puncture to 88.9% (8/9) with robotic assistance. These results suggest the preliminary effectiveness of the proposed shared-control robotic system under dynamic RM conditions, although further larger-sample validation is still required.

Labrador dog in vivo experimental validation

To verify the feasibility and stability of the proposed shared-control puncture system under realistic physiological respiratory conditions, an in vivo animal experiment was conducted. All animal experiments in this study were reviewed and approved by the Animal Welfare Ethics Committee of Shanghai Sixth People's Hospital (animal experiment registration No. DWSY2025-0100). The experiments were performed in the Animal Laboratory of Shanghai Sixth People's Hospital (facility license No. SYXK (Hu) 2021-0028), and all methods were carried out in accordance with the relevant institutional guidelines and regulations for animal welfare and experimental procedures. An adult male Labrador dog, approximately 60 months old and weighing about 32.5 kg, was used in this study. The animal was supplied by Shanghai Jiagan Biotechnology Co., Ltd. (animal production license No. SCXK (Hu) 2025-0007). The animal was housed in the animal facility and underwent standardized preoperative observation and care for 28 days prior to the experiment. Before the procedure, food was withheld for 12 h while water was allowed ad libitum. A peripheral intravenous catheter was placed in the right forelimb, and atropine sulfate (0.02 mg/kg, i.v.) was administered before anesthesia. General anesthesia was induced with propofol (2.5 mg/kg, i.v.) and maintained by continuous intravenous propofol infusion (20–25 mg/kg/h). After induction, the dog was endotracheally intubated and mechanically ventilated. Heart rate, respiratory rate, oxygen saturation, and noninvasive blood pressure were continuously monitored throughout the procedure, and supplemental propofol was administered when necessary. Informed consent was obtained from the identifiable individual shown in Fig. 8 for publication of the image in an online open-access publication. During the experimental preparation stage, a 2% (w/w) sodium alginate solution was implanted into the liver to serve as a tumor-mimicking marker, thereby constructing a target region that could be stably identified in US imaging. The overall

experimental procedure is illustrated in Fig. 8c–g. First, four retroreflective markers were attached to the animal's body surface to facilitate respiratory phase estimation and establish a spatial reference frame. Subsequently, the animal was transferred to the CT suite for preoperative scanning, and the acquired imaging data were imported into the SCC to complete 3D reconstruction of the liver and target region, as shown in Fig. 8c. Based on the reconstructed anatomical information, the surgeon planned the target body-surface pose of the probe according to clinical experience, aiming to obtain a clear US imaging view covering the tumor region. Meanwhile, the robotic manipulator carrying the PEM with the probe was guided to move toward the planned body-surface region along the predefined path, as illustrated in Fig. 8d. Once the probe established contact with the skin surface, the system switched to a force-regulated assistance mode to maintain a stable probe-tissue contact force, thereby suppressing imaging disturbances induced by RM. On this basis, the surgeon fine-tuned the probe pose according to real-time imaging feedback, such that the target tumor was stably visualized within the imaging plane, and subsequently performed the puncture along the planned needle trajectory, as shown in Fig. 8e. Under the stabilized imaging condition provided by the force-regulated assistance, the surgeon selected an appropriate timing to complete the puncture based on real-time US feedback. The puncture outcome is presented in Fig. 8f. After the procedure, the tissue specimen obtained through the puncture needle was extracted and placed on sampling paper for inspection, as shown in Fig. 8g. This single-animal experiment served as an in vivo feasibility demonstration, showing that the proposed shared-control puncture system could support surgeon-performed puncture procedures. However, further repeated in vivo experiments with quantitative targeting metrics are required to comprehensively evaluate its robustness and translational potential.

Conclusion and discussion

The shared-control US-guided liver puncture robotic system is proposed to address the challenges introduced by RM during clinical puncture surgeries. The PEM is specifically designed to enable concurrent US probe manipulation and needle handling, while preserving the surgeon's authority during needle orientation and insertion. In addition, a dual-stage pose–force assistance scheme is implemented. The pose-guided probe positioning stage establishes a favorable and spatially consistent imaging configuration, whereas the force-regulated interaction stabilization stage maintains stable probe–skin contact during RM. Together, these components form a surgeon-in-the-loop shared-control paradigm that enhances puncture procedural robustness without disrupting conventional clinical workflows.

Comprehensive experimental evaluations are conducted to validate both individual modules and overall system performance. The BRMD is developed as a reproducible experimental instrument and platform, capable of realistically mimicking thoracoabdominal anatomy and RM using heterogeneous, medical-imaging-compatible materials. Systematic experiments are performed to verify probe positioning accuracy and force-regulated interaction stability under RM. Phantom-based puncture experiments on the BRMD further demonstrate that the proposed shared-control strategy improves procedural efficiency, accuracy, and success rate compared with manual operation. In addition, an in vivo Labrador dog experiment is conducted under physiological RM to evaluate robustness and effectiveness in the presence of realistic tissue properties and respiratory variability, which can support its potential for clinical translation.

Despite these results, the current validation still has several limitations that should be addressed in future work. The phantom and in vivo experiments were mainly designed to evaluate the feasibility and preliminary effectiveness of the proposed shared-control system, and the sample size remains limited.In addition, because operator experience and operating habits may influence system performance, further validation involving more operators is helpful. The in vivo Labrador dog experiment served as a qualitative feasibility demonstration under physiological RM, while quantitative in vivo targeting metrics, such as needle-to-target error, entry-point deviation, and targeting success rate, were not systematically measured. Future work will therefore include larger-sample validation, multi-operator studies, quantitative in vivo targeting assessment, and the incorporation of US image processing and clinical operation data to further enhance the perceptual and assistive capabilities of the shared-control system.

Data availability

The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

Received: 31 March 2026; Accepted: 28 May 2026
Published online: 08 June 2026

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Author contributions

W.Z. conceived the study, conducted the experiments, analysed the results, and drafted the manuscript. W.L., W.Y., and L.X. assisted with the experiments, data collection, and result analysis. Q.L. provided clinical support and experimental assistance. Y.Y. supervised the study, provided clinical guidance, and served as the corresponding author. All authors reviewed and approved the manuscript.

Funding

This work was supported by the National Key R&D Program of China (Grant Nos. 2022YFC2405503 and 2022YFC2405500) and the National Natural Science Foundation of China (Grant Nos. W2521022 and 62473258).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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